

VTU Belagavi

SUPPLEMENTARY EXAMINATION ? 2025

Analog and Digital Communications ? Supplementary Examination, 2025

Subject Code: EC401 Semester: 4 Regulation: R18

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Amplitude Modulation)

1. Define Amplitude Modulation. Explain its significance in Analog and Digital Communications with a suitable example. [12 Marks]
2. Write short notes on Amplitude Modulation. [12 Marks]
3. Write short notes on Amplitude Modulation. [12 Marks]

SECTION B (Unit 2: Angle Modulation)

4. Derive the expression related to Angle Modulation and explain each term. [8 Marks]
5. Explain Angle Modulation with a neat diagram/sketch and a suitable example. [12 Marks]
6. Compare and contrast the key concepts involved in Angle Modulation. [12 Marks]

SECTION C (Unit 3: Digital Modulation Techniques)

7. Compare and contrast the key concepts involved in Digital Modulation Techniques. [12 Marks]
8. Derive the expression related to Digital Modulation Techniques and explain each term. [12 Marks]
9. Define Digital Modulation Techniques. Explain its significance in Analog and Digital Communications with a suitable example. [12 Marks]

SECTION D (Unit 4: Information Theory and Coding)

10. Write short notes on Information Theory and Coding. [8 Marks]
11. How is Information Theory and Coding implemented? Give an algorithm / procedure and explain. [8 Marks]
12. Describe the working principle of Information Theory and Coding with necessary equations. [8 Marks]

SECTION E (Unit 5: Multiplexing and Multiple Access)

13. Solve a problem related to Multiplexing and Multiple Access. Show all the steps clearly. [12 Marks]
14. List the advantages and limitations of Multiplexing and Multiple Access. Explain any two in detail. [12 Marks]
15. Explain Multiplexing and Multiple Access with a neat diagram/sketch and a suitable

example. [12 Marks]

*** End of Question Paper ***